CHAPTER

17

Reevaluating the role of the STN-GPe network in generating beta synchronized oscillatory activity in parkinsonism

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1. From normal to abnormal BG dynamics following the loss of dopamine in parkinsonism

BG circuits play a crucial role in learning, selecting, and optimizing actions that lead to rewarding outcomes. Early models suggested that these functions are mediated through firing rate changes across BG networks, starting at the level of the striatum and ultimately leading to motor program disinhibition in the thalamo-cortical system [1,2]. Such "rate-coding models" have been highly influential in shaping our understanding of the anatomo-functional organization of BG circuits (Fig. 17.1A), and the behavioral consequences of firing rate changes in BG have been demonstrated experimentally. Indeed, selective opto- or pharmacogenetic manipulation of striatal direct or indirect neurons in the motor loop profoundly impacts movement, facilitating or inhibiting motoric aspects, respectively [4–6].

However, a major limitation of this "rate-coding" framework is that it does not account for other neuronal dynamics, such as synchronized oscillatory activity, which is a fundamental feature of brain circuits [7]. In particular, accumulating evidence suggests that, under physiological conditions, BG circuits express brief burst (<200 ms) of synchronized beta oscillations that are precisely timed-locked to task-relevant events (e.g., cue presentation or movement completion) and play a role in neural operation [8–14].

The loss of dopamine in PD profoundly alters both firing rate and synchronized oscillatory dynamics in BG (Fig. 17.1B). Rate modifications include hyperactivity of the indirect D2-expressing striatal projection neurons (D2-SPNs) and hypoactivity of the direct D1-expressing striatal projection neurons (D1-SPNs) [15,16], although these changes in activity depend on the motor state—for example, during locomotion versus immobility [17,18]. Together, these rate modification are thought to lead to an excessive inhibition of motor programs via increased BG GABAergic output to the thalamo-cortical and brainstem motor regions. However, experimental data supporting this BG output hyperactivity remain conflicting [19–23,23a]. In parallel to these rate changes, dopamine depletion also disrupts the expression of beta oscillations, shifting them from a highly transient and dynamic nature to more prolonged and powerful beta bursts, as most prominently observed in PD patients [24]. This change in beta oscillations dynamics contributes to excessive spike synchronization at beta frequencies in BG circuits [25,26].

Although beta synchronization is strongly correlated with PD motor symptoms, accumulating evidence suggests a temporal dissociation between the onset of akinetic symptoms and the expression of pathological beta oscillations. Specifically, motor deficits in PD appear before the emergence of synchronized beta activity, as observed in MPTP-treated monkeys [27] and in toxin-based (6-OHDA) or pharmacological rat models of PD [28–31]. Importantly, this chapter will not focus on the precise causal contributions of firing rate changes versus abnormal synchronization to PD motor

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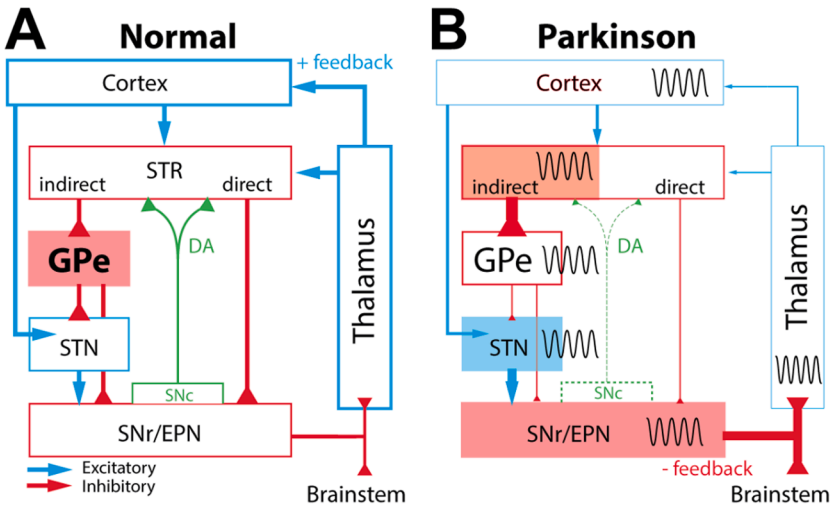


FIGURE 17.1 Firing rate versus Synchronization in classic basal ganglia model. (A–B), Schematic representation of the classic "rate based" basal ganglia model in normal condition (A) or in Parkinson's disease (B) [1, 2]. Abnormal spike synchronization at beta frequency in BG circuits were only detected following the loss of dopamine in PD [3]. Abbreviations are STR: Striatum; GPe: Globus Pallidus external part; STN: Subthalamic nucleus; SNr: Substantia nigra pars reticulata; EPN: Entopedoncular nucleus.

symptoms, as these topics are explored in greater detail elsewhere in this handbook (see Chapter 14 by Qiaoling et al., Chapter 19 by Judith Walters and Petersson, and Chapter 21 by Aman et al.). Rather, we will examine the neuronal substrates within the BG that contribute to the generation and propagation of the parkinsonian beta oscillations in 6-OHDA unilaterally lesioned rat model of PD. While alterations in firing rate dynamics within the BG likely stem from the striatum as a result of striatal dopamine depletion [20,32], the mechanisms driving the abnormal increase in beta synchrony remain poorly understood.

2. Synchronized beta oscillations in BG neuronal networks

Any closed biological system that incorporates a delayed negative feedback loop inherently possesses rhythmic properties, including the ability to generate oscillations in response to external perturbations [33]. This principle also applies to neuronal networks integrating negative feedback mechanisms, a basic example being a reciprocally connected network of inhibitory and excitatory neurons [34]. Following small external perturbations, such network can generate oscillations that may be either transient or persistent, depending on the system's dynamical regime (e.g., stable stationary vs. persistent oscillatory activity). In the case of transient oscillations, the rhythmic activity decays over time and eventually returns to the system's stable steady state (Fig. 17.2A). However, changes in key parameters—such as the strength of the positive or negative feedback, the strength of mutual coupling within nodes, synaptic delays, and membrane time constants, among other relevant factors—can push the system past a critical threshold, leading to the generation of more sustained oscillations (Fig. 17.2B).

The complex anatomo-functional organization of BG neuronal networks, along with their projections to thalamo-cortical systems, forms multiple parallel and recurrent inhibitory feedback loops (Fig. 17.2C) [36–38], many of which represent putative candidates for generating transient oscillatory activity such as the beta transients observed in physiological conditions [39]. In PD, however, dopamine loss triggers widespread functional alterations, profoundly affecting the dynamical properties of these circuits. Some of these changes might first occur in the striatum following dopamine loss. For example, lack of D2 receptor stimulation has been associated to stronger beta oscillations expression [40], while increased activity of striatal cholinergic interneurons can enhance beta oscillations in healthy mice and induce PD-like motor deficits [41]. Similar functional or anatomical changes have also been reported throughout many regions of BG circuits under PD conditions [42–47]. These alterations—whether compensatory, pathological, or incidental consequences of dopamine depletion—can strongly alter the gain of specific negative feedback loops, potentially shifting them from a transient beta generation mode to a more persistent state of beta-activity, thereby contributing to excessive synchronization within these networks [48].

While the precise circuit mechanisms underlying excessive beta oscillations in animal model of PD remain unclear, numerous studies have implicated the motor cortex, the striatum, the STN, and the GPe in various aspects of beta oscillation dynamics, including their generation, amplification, and propagation [19,49–51, 51a]. In addition, a key feature of beta oscillations is the precise spike-timing relationship they impose on BG neurons, giving rise to specific phase alignments between

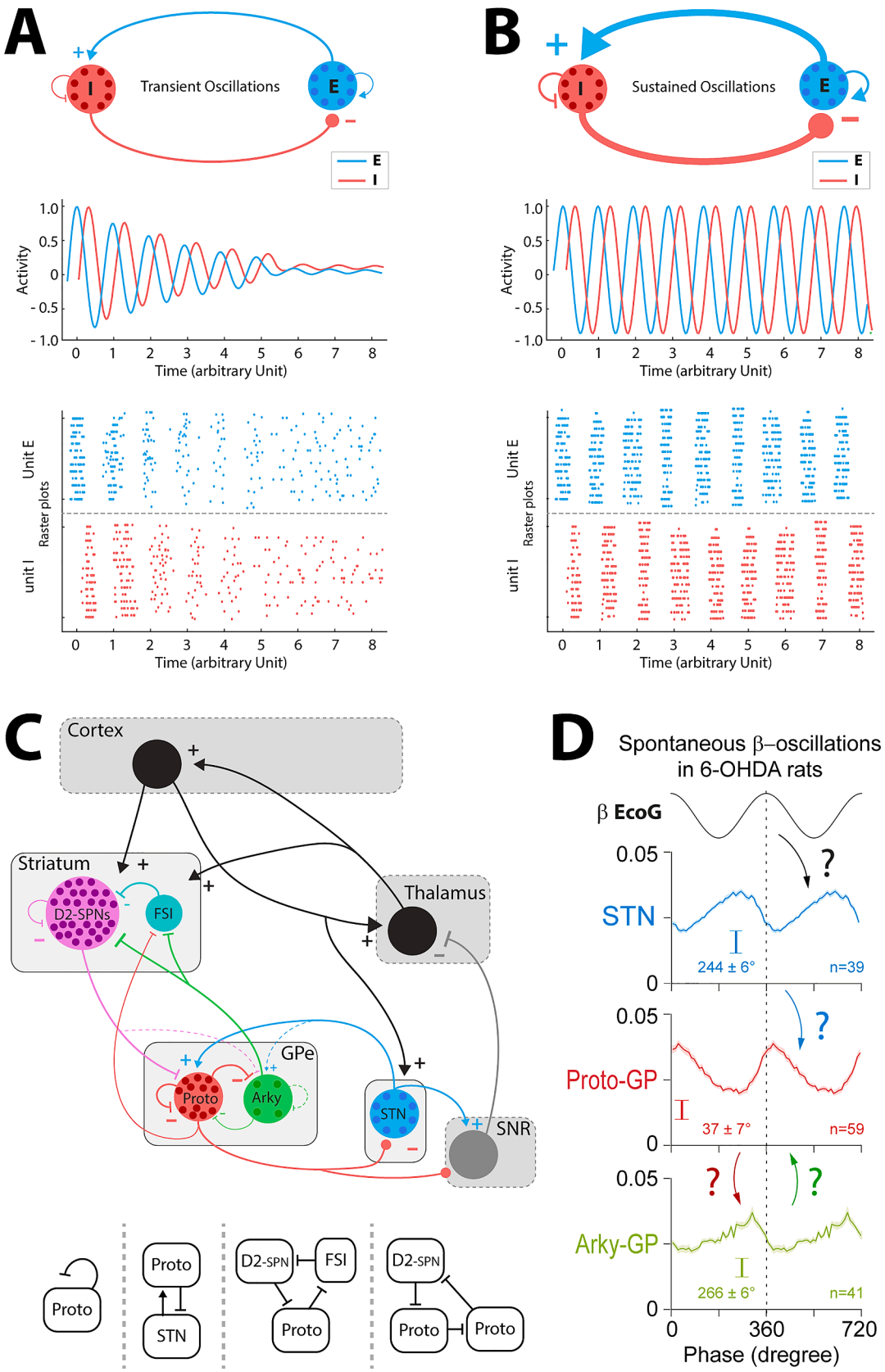


FIGURE 17.2 **From theory to practice: synchronized beta oscillatory activity in basal ganglia.** (A–B) Schematic representation of a simple reciprocally connected inhibitory/excitatory (I/E) feedback neuronal network (*top*), along with theoretical mean activity plots (*middle*) and raster plots (*bottom*) for the two sets of neurons (red: inhibitory, blue: excitatory). Panel A shows the condition in which oscillatory activity is transient,

subpopulation of neurons that has been particularly well characterized for STN-GPe neurons in 6-OHDA rat PD model (Fig. 17.2D). As such, a prominent hypothesis suggests that beta oscillations originate in the cortex and its loop with the STN via the hyperdirect pathway [36,52], where they could get abnormally amplified through recurrent STN–GPe feedback mechanism [53].

That being said, the STN-GPe negative feedback loop has been proposed to act as a BG central pacemaker for the generation of synchronized oscillatory activity, even in the absence of cortical inputs. Indeed, this network can exhibit rhythmic properties in organotypic slices [54]. The synaptic organization and intrinsic properties of STN and GPe neurons can further reinforce the recurrent feedback properties of this circuit. Anatomically, GPe projections to STN are sparse yet highly selective and potent, predominantly targeting the proximal dendrites and soma of STN neurons [54a]. Studies conducted in brain slices have demonstrated that activation of just a few GPe inputs is sufficient to inhibit, reset, and synchronize the autonomous activity of STN neurons [54a,b,c]. In addition, both STN and GPe neurons express voltage-gated ion channels that confer intrinsic pacemaking properties, allowing these neurons to generate spontaneous rhythmic activity independent of excitatory synaptic inputs [54d,e]. However, while it is true that STN and GPe neurons exhibit strong pacemaking properties at the single-cell level, the STN-GPe network as a whole does not appear to generate strong synchronized spike-timing activity under physiological conditions [35,55,56]. Instead, it displays low levels of oscillatory activity spike cross-correlation, likely due to active decorrelation mechanisms present in BG network [56a,b]. Indeed, when examining entrainment between spikes and the mean field (i.e., local field potential), Leventhal et al. found that only a small fraction of BG neurons (<10%) were phase-locked to beta oscillations in healthy animals engaged in a motor task [9]. However, following dopamine depletion, synaptic and circuit-level adaptations may disrupt the balance of excitation and inhibition within BG circuits, in particular the STN-GPe network. This disruption may impair active decorrelation mechanisms, shifting the network toward an abnormally exaggerated state of synchronization, as evidenced by the strong increase in beta coherence—reaching up to 44%—between pairs of GPe neurons [35].

3. An update of the STN-GPe microcircuit: cellular diversity and functional connectivity

The STN-GPe network has traditionally been considered functionally homogenous, with all GPe neurons projecting to STN, and vice versa. However, this monolithic view is both oversimplified and inaccurate, especially as it fails to account for the cellular diversity present within this network and the potential functional connectivity enabled by this diversity. While some cell diversity seems to exist in the STN [57,58], it has been particularly well characterized in GPe neurons across multiple functional levels: molecular [59], embryonic origin [60], structural [61–63], electrophysiological [64–67], and behavioral [68–70]. Our work has also revealed that the GPe functional heterogeneity becomes evident in the 6-OHDA lesioned rats. Indeed, while most GPe neurons exhibit tonic firing and remain uncorrelated in control rats, dopamine depletion reveals two distinct population of GPe neurons, each strongly correlated with both slow-wave or beta oscillations but with opposite spike-timing phase relationships [35,61]. These results ultimately led to the discovery of the dichotomous organization of GPe neurons based on in vivo juxtacellular labeling to correlate the electrophysiological, with molecular/structural properties of individual GPe neurons [61]. As such, we found that the predominant GPe cell-type, representing approximately 70% of all cells, corresponds to prototypic neurons that consistently project long-range axons to the STN and all major downstream BG nuclei (Fig. 17.3A). This canonical population originates developmentally from the medial ganglionic eminence [73] and is characterized by the expression of molecular markers such as parvalbumin (expressed in the majority, but not all, of these cells), as well as transcription factors including NK2 homeobox 1 (Nkx2.1) and LIM homeobox 6 (Lhx6) [73–76]. More recently, the classification of prototypic neurons has been further divided into molecularly defined subclasses, which are likely to contribute to distinct behavioral specializations [63,76,77]. The remaining

while panel B illustrates the scenario where modifications to the connectivity/coupling strength induce sustained activity. (C) Simplified schematic of the BG in the parkinsonian state, illustrating the main inter-nuclei connections (*top*), and example of four neuronal circuits integrating negative feedback loops (*bottom*). The direct D1-expressing striatal projection neurons are not represented due to their decreased activity in the PD state [15,17]. (D) Mean phase histograms of STN-GPe neurons illustrating the spike-timing phase relationship during parkinsonian beta oscillations, along with a proposed scenario (arrows) for their generation and propagation within cortico-BG circuits. Vertical lines under phase histograms are means ± 2 SEM. Abbreviations are: GPe: Globus Pallidus external part; STN: Subthalamic nucleus; SNR: Substantia nigra pars reticulata. *Figures panel D was modified from N. Mallet, A. Pogosyan, L.F. Márton, J.P. Bolam, P. Brown, P.J. Magill, Parkinsonian beta oscillations in the external globus pallidus and their relationship with subthalamic nucleus activity. J. Neurosci. 28 (2008a) 14245–14258.*

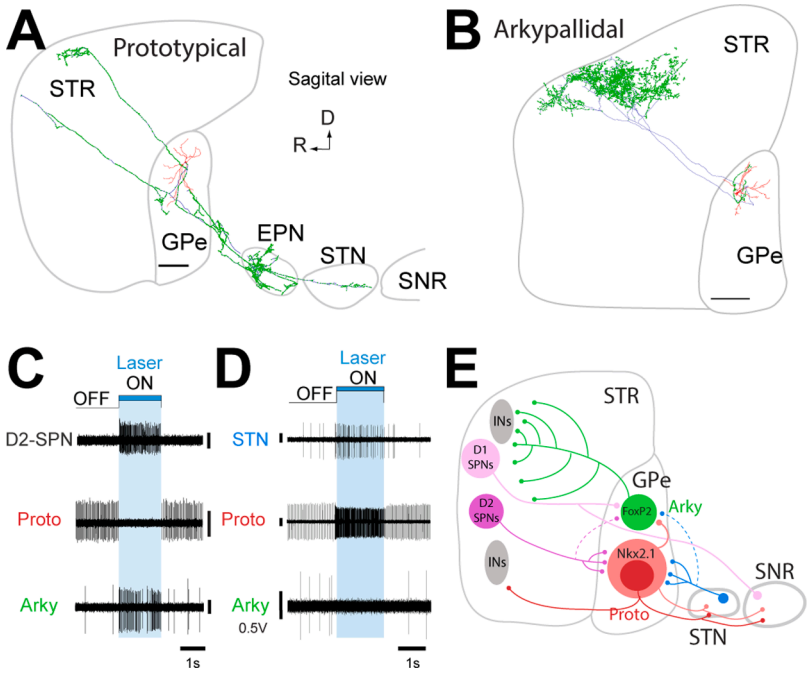


FIGURE 17.3 **Prototypic and arkypallidal structural properties and functional inputs organization.** A,B, Full reconstruction of one prototypic (Proto, A) and one arkypallidal (Arky, B) GPe neuron view in sagittal plane. Somata and dendrites are drawn in red, axons in blue, and axonal boutons in green. Note that this prototypic neuron belongs to the one-third subpopulation of prototypic neuron that project back to striatum [71]. C,D, Representative examples illustrating the in vivo impact of D2-SPNs (C) and STN (D) opto-excitation on prototypic and arkypallidal neurons. Vertical Scale bars represent 0.5 mV. The optogenetic strategy relied on D2-Cre (C) or Vglut2-Cre (D) mice with AAV-DIO-ChR2-eYFP in the striatum or STN, respectively. E, Schematic illustrating the functional connectivity of D2-SPNs, D1-SPNs, and STN inputs onto prototypic and arkypallidal neurons. Abbreviations are: Arky: arkypallidal GPe neuron; D1-SPNs: D1-expressing striatal projection neurons; D2-SPNs: D2-expressing striatal projection neurons; GPe: external part of the Globus Pallidus; Ins: striatal interneurons; Proto: prototypic GPe neuron; SNR: Substantia nigra pars reticulata, STN: Subthalamic nucleus; STR: striatum. *Figure panels A–B were modified from N. Mallet, B.R. Micklem, P. Henny, M.T. Brown, C. Williams, J.P. Bolam, K.C. Nakamura, P.J. Magill, Dichotomous organization of the external globus pallidus. Neuron. 10974199. 74 (6) (2012), 1075–1086. https://doi.org/10.1016/j.neuron.2012.04.027. Figure panels C–D were extracted from A. Aristieta, M. Barresi, S.A. Lindi, G. Barrière, G. Courtand, B. de la Crompe, L. Guilhemsang, S. Gauthier, Stéphanie F., Jérôme B., N.P. Mallet, A disynaptic circuit in the globus pallidus controls locomotion inhibition. Curr. Biol. 31 (4) (2021), 707. https://doi.org/10.1016/j.cub.2020.11.019.*

population of GPe cells, known as arkypallidal neurons, project exclusively to the striatum, where they provide a major source of extrinsic GABAergic input [61] (Fig. 17.3B). If we first identified this GPe dichotomous organization in parkinsonian animals, its existence was later confirmed in control rats [75] and mice [73]. Unlike prototypic neurons, arkypallidal cells originate from the lateral ganglionic eminence and are defined by a distinct molecular profile, including the opioid precursor preproenkephalin and the transcription factors Forkhead box protein P2 (FoxP2) and Meis homeobox 2 (Meis2) [61,73–75]. In conclusion, the projections from arkypallidal neurons reveal a novel mode of striatal GABAergic inhibition, although the operational principles of this control remain largely unknown [78].

As part of the indirect pathway, the GPe receives most of its inputs from two main BG structures of opposite nature: GABAergic inputs from striatal spiny neurons (particularly D2-SPNs) and glutamatergic afferents from STN. A key question that comes to mind is whether the dichotomous organization of GPe neurons extends to their input connectivity, or whether D2-SPN and STN inputs impact prototypic and arkypallidal neurons equally. This functional connectivity has been directly investigated using optogenetic approaches in D2-Cre and Vglut2-Cre mice, enabling selective manipulation of D2-SPNs and STN neurons, respectively [72,79]. These experiments demonstrate that GPe neurons integrate inputs from both D2-SPNs and STN neurons in a cell-type-specific manner. Specifically, both D2-SPN and STN inputs preferentially impact prototypic neurons, resulting in opposite response patterns between prototypic and arkypallidal neurons (Fig. 17.3C–D). The opposition of activity between both GPe neurons arises from the powerful inhibitory control that prototypic neurons exert on arkypallidal neurons through their dense axon collaterals [80]. This mirrored activity resembles the opposite spike-timing phase relationship observed during slow wave and beta oscillations in parkinsonian rats, as discussed earlier [61].

Accordingly, in vivo optogenetic excitation of D2-SPNs inhibits prototypic neuron activity, thereby

disinhibiting arkypallidal neurons and increasing their firing rate (Fig. 17.3C). This difference in effect was caused by the fact that the magnitude of D2-SPNs inhibitory inputs measured in slice recordings was three times bigger in prototypic compared with arkypallidal neurons [72]. Under these conditions, the reduced inhibition from prototypic neurons also leads to the disinhibition of STN neurons, explaining their concurrent increase in firing activity. Conversely, in vivo optogenetic stimulation of the STN preferentially excited prototypic neurons, leading to disynaptic inhibition of arkypallidal neurons (Fig. 17.3D). Once again, this effect could be explained by the fact that STN inputs in vitro were 74% smaller in arkypallidal than prototypic neurons, while the opposite activity between prototypic and arkypallidal neurons strongly suggests the involvement of collateral inhibition [72]. It is important to mention that this differential input organization also applies to collateral projections from D1-SPNs to the GPe, which preferentially target arkypallidal neurons [79]. Altogether, these results call for a revaluation of how the direct/indirect, and hyperdirect pathway interact at the level of GPe neurons (Fig. 17.3E). Moreover, these functional connectivity rules are not only crucial for understanding normal GPe motor function but are likely fundamental in governing how synchronized activity, such as parkinsonian beta oscillations, emerge and propagate within the STN-GPe network.

4. Optogenetic interrogation of the BG mechanism underlying the generation of parkinsonian beta oscillations

The mechanism underlying parkinsonian neuronal oscillations has been investigated using pharmacological manipulations, such as local infusion of GABAergic or glutamatergic agonist and antagonist into the GPe or STN, in nonhuman primates BG [51,81]. This work has highlighted the role of cortical glutamatergic inputs to the STN and the reciprocal connection between the STN and GPe in propagating and amplifying oscillatory activity, which occurred in the low beta frequency range (8–15 Hz). This raises the question of whether similar mechanisms are at play in parkinsonian rodents, where beta oscillations occur at faster frequencies (~20 Hz).

The development of circuit-specific optogenetic manipulations in rodents enables direct dissection of the contributions of different BG circuit components and testing of hypotheses related to the generation and propagation of parkinsonian beta oscillations. Notably, compared with pharmacological manipulations, optogenetic approaches provide a powerful tool adapted to the fast dynamics of beta oscillations in vivo, as they allow millisecond timescale ON/OFF control of neuronal activity. Besides, optogenetic stimulations can be combined with electrophysiological recordings in parkinsonian rats for a fast readout of both the effects and their recovery [82].

Given the potential roles of STN, and GPe in the generation/propagation mechanisms of beta oscillations, we systematically assess their involvement using an optogenetic inhibition strategy in parkinsonian rats. This approach was based on the expression of Archaerhodopsin-3 (ArchT3), a light-activated outward proton pump that hyperpolarizes and inhibits neurons under green/yellow light stimulation [83]. We also tested whether beta oscillations depended on motor cortex (mCx) activity like in nonhuman primates. Indeed, these oscillations are readily detected in electrocorticogram (ECoG) recordings from this region in 6-OHDA rats [31,84], and cortical circuits are well known to generate oscillatory activity across multiple frequency bands [85]. Viral-mediated ArchT3 expression in mCx, STN, or GPe resulted in robust neuronal inhibition upon yellow light delivery, as confirmed by electrophysiological recordings [82]. However, power spectral analysis of ECoG recordings revealed distinct effects: neither mCx (Fig. 17.4A), nor STN opto-inhibition (Fig. 17.4B), significantly altered beta oscillation power, whereas GPe inhibition led to a pronounced suppression of beta power (Fig. 17.4C).

Altogether, these findings strongly refute the involvement of mCx or STN in generating beta oscillations while supporting the hypothesis that GPe activity is necessary for beta synchronization in 6-OHDA rats. This suggests that the GPe is a key driver of abnormal oscillations in BG circuits, and that beta oscillations do not propagate to the GPe via excitatory inputs from the STN, as proposed in nonhuman primate BG.

To further elucidate the circuit mechanisms underlying parkinsonian beta oscillations and to validate our previous findings, we developed an alternative optogenetic approach by reintroducing synthetic beta oscillations through direct patterning of GPe (Fig. 17.5A) or STN (Fig. 17.5B) neuronal activities using light stimulation. This stimulation mimicked both the frequency (20 Hz) and sine-wave oscillatory pattern of beta oscillations. These two approaches were based on distinct optogenetic strategies: in the GPe, we applied optogenetic inhibition to examine the effects of rhythmic inhibitory input on beta oscillations properties, whereas in the STN, we used optogenetic excitation to determine the impact of rhythmic excitatory inputs to STN (such as those coming from mCx) and how they would propagate to GPe neurons.

To determine which beta-patterned optogenetic protocol most accurately replicated the properties of real parkinsonian beta oscillations, we compared the spike-timing, phase properties, and firing rate changes

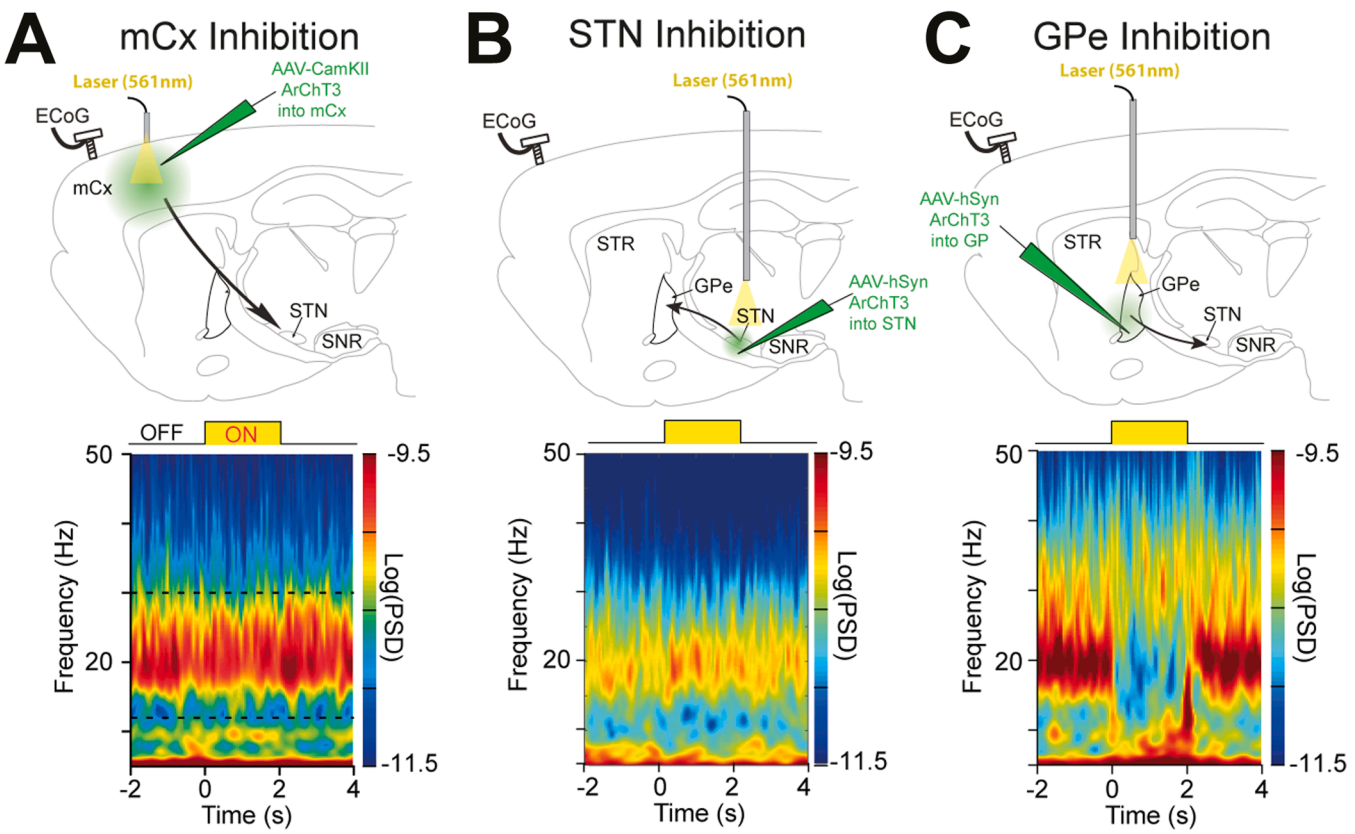


FIGURE 17.4 **Optogenetic mapping of parkinsonian beta oscillations dynamics in basal ganglia.** A–C, Schematic of the experimental paradigm (*top*) and peri-event spectrogram (*bottom*) of mCx electrocorticogram (ECoG) recording showing the beta-frequency power during mCx (A), STN (B), and GPe (C) opto-inhibition. Abbreviations are: ECoG: electrocorticogram; GPe: external part of the Globus Pallidus; mCx: motor cortex; SNR: Substantia nigra pars reticulata, STN: Subthalamic nucleus; STR: striatum. *Figure panels A–C were extracted from B.d.l. Crompe, A. Aristieta, A. Leblois, S. Elsherbiny, T. Boraud, N.P. Mallet, The globus pallidus orchestrates abnormal network dynamics in a model of Parkinsonism. Nature Res. France Nat. Commun. 11 (1) (2020) 20411723, Available from: https://doi.org/10.1038/s41467-020-15352-3*

induced by these synthetic beta oscillations in the STN-GPe network to those of spontaneous beta oscillations (Fig. 17.5 C and D). Consistent with our hypothesis, optogenetic inhibition of GPe neurons best replicated key features of parkinsonian beta oscillations, including the correct phase-locking properties of STN and prototypic neurons, as well as the characteristic anti-phase relationship between them. Regarding spike-rate changes, this protocol successfully reproduced STN hyperactivity and prototypic hypoactivity as in parkinsonian animals but failed to replicate the opposing activity patterns of prototypic and arkypallidal neurons—likely because optogenetic inhibition affected both cell types equally. In contrast, optogenetic excitation of the STN at beta frequency induced an abnormal phase-locking of STN and prototypic neurons, an incorrect in-phase relationship between them, and an aberrant increase in prototypic neuron firing rates. However, some aspects of parkinsonian beta oscillations were preserved, such as the STN hyperactivity and the phase opposition between prototypic and arkypallidal neurons, which arises from the differential synaptic organization of STN inputs as discussed in the previous section. Altogether, these results confirm that the generation of abnormal beta synchronization in 6-OHDA rodent is not driven by STN excitatory inputs (unlike in nonhuman primates) [81,86], but rather by increased inhibitory input to GPe prototypic neurons. This inhibitory input may originate from striatal D2-SPNs, which are selectively entrained to parkinsonian beta oscillations [16] and, through their preferential connectivity to prototypic neurons [72], could underlie the observed phase opposition between prototypic and arkypallidal neurons. The previous findings also underscore the pivotal role of prototypic neurons in shaping firing rate dynamics and orchestrating neuronal synchronization within the STN. While the minimal neuronal circuits required for the generation of beta oscillations remain unknown, striato-pallidal loops have been implicated [87,88]. In addition, broader circuits—including

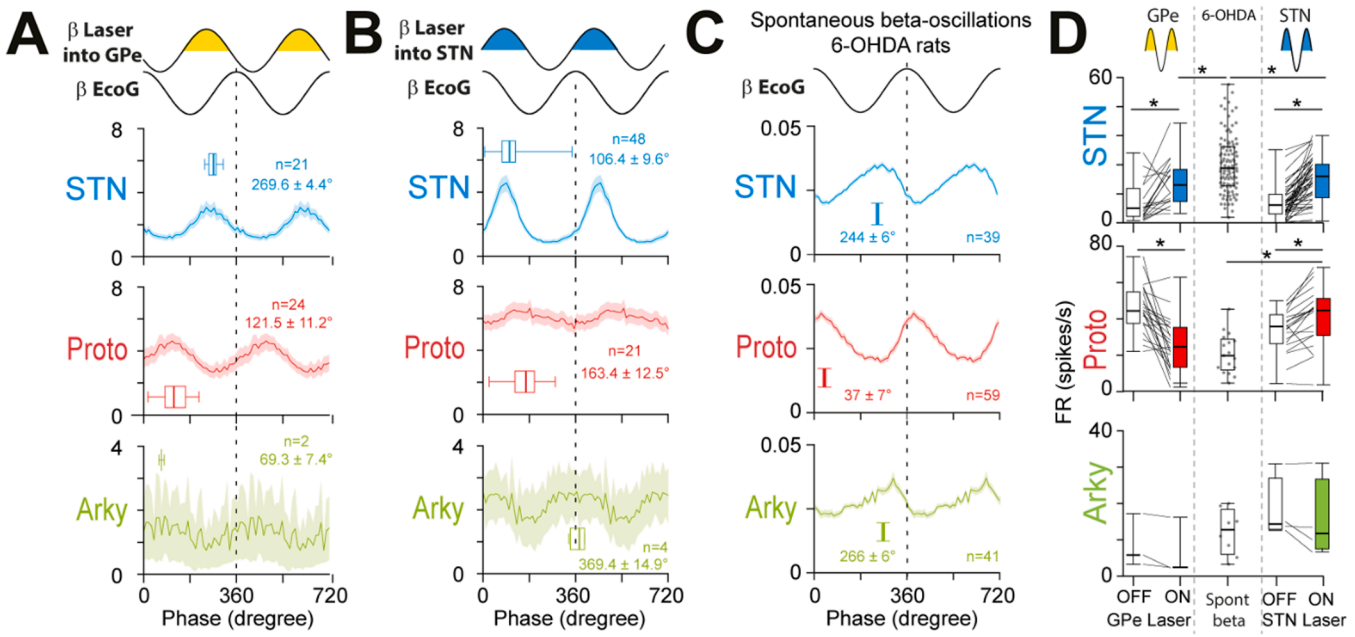


FIGURE 17.5 **Spike-timing and spike-rate properties of spontaneous or opto-evoked beta oscillations in STN-GPe network.** A–C, Mean phase histograms of STN (*top*, blue), prototypic (*middle*, red), and arkypallidal (*bottom*, green) neurons during spontaneous parkinsonian beta oscillations recorded in 6-OHDA-lesioned rats (A), or synthetic beta oscillations evoked by beta-frequency opto-patterning of GPe (B) or STN (C) neurons in control rats. The beta signal recorded from the motor cortex ECoG was used as the reference signal to calculate the linear phase histograms for the different neuronal populations. D, Comparison of the spike-rate changes in STN (*top*, blue), prototypic (*middle*, red), and arkypallidal (*bottom*, green) neurons during spontaneous (Spont β, middle) or opto-evoked GPe (GPe Laser, *left*) or STN (STN laser, *right*) beta oscillations. *Figure panels A, B, and D were extracted from B.d.l. Crompe, A. Aristieta, A. Leblois, S. Elsherbiny, T. Boraud, N.P. Mallet, The globus pallidus orchestrates abnormal network dynamics in a model of Parkinsonism. Nat. Res. France Nat. Commun. 11 (1) (2020) 20411723. https://doi.org/10.1038/s41467-020-15352-3. Figure panels C was extracted from N. Mallet, A. Pogosyan, L.F. Márton, J.P. Bolam, P. Brown, P.J. Magill, Parkinsonian beta oscillations in the external globus pallidus and their relationship with subthalamic nucleus activity. J. Neurosci. 28 (2008a) 14245–14258, N. Mallet, A. Pogosyan, L.F. Márton, J. P. Bolam, P. Brown, P.J. Magill Disrupted dopamine transmission and the emergence of exaggerated beta oscillations in subthalamic nucleus and cerebral cortex. J. Neurosci. 28 (2008b) 4795–4806.*

projections from basal ganglia output nuclei such as the substantia nigra pars reticulata to the thalamus—may also play a role [89].

5. Comparison of the spike-timing properties of parkinsonian beta oscillations and spike-and-wave discharges

Absence seizures, a hallmark of certain types of non-convulsing epileptic syndromes characterized by transient loss of consciousness and typically developing during brain maturation in childhood, are associated with the abnormal expression of synchronous spike-and-wave discharges (SWDs) on the electroencephalogram. Genetic rat models of absence epilepsy, such as the GAERS rats, replicate the expression of SWDs and provide a valuable model for investigating their mechanisms of generation [90]. Notably, SWDs propagate to BG circuits, inducing pronounced spike synchronization within the STN-GPe network (Fig. 17.6A) [91]. These oscillations occur at a lower frequency (7–11 Hz) than beta oscillations, are independent of dopamine loss, and depend on the integrity of both cortical and thalamic circuits [90,92]. The fact that cortical inputs are necessary for the generation of SWDs but appear dispensable for beta oscillations [82] suggests that distinct mechanisms underlie their generation. Despite these fundamental differences, analyzing the spike-timing phase relationships induced by both rhythms in the STN-GPe network can yield key insights into the neuronal mechanisms driving the propagation of excessive synchronization in BG circuits. Based on their known firing entrainment [91], we extrapolated the linear phase histograms of STN-/GPe neurons during SWDs (Fig. 17.6B), to best compare their spike-timing phase properties with those observed during beta oscillations (Fig. 17.5C).

A striking finding from Ref. [91] was that GPe recordings during SWDs also revealed two distinct cell population based on firing patterns. Although the prototypic and arkypallidal classification had not yet been established at the time, the authors identified a subset of GPe neurons, named “early discharge” (ED-GP) neurons, comprising ~25% of the recorded GPe population

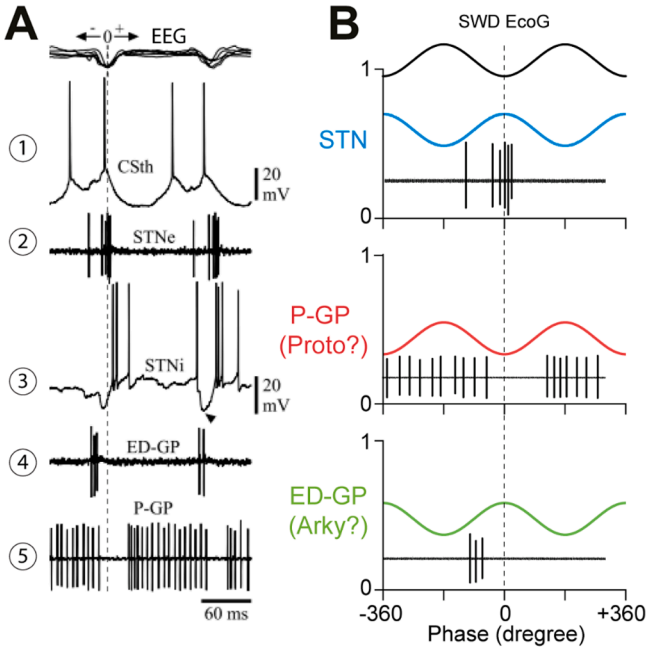


FIGURE 17.6 **Spike-timing relationship of STN-GPe neurons during spike-and-wave discharges oscillations.** (A) Representative example illustrating two successive spike-wave discharges recorded in the EEG (top traces) and corresponding intracellular activity of cortical neurons (1), extracellular (2) and intracellular (3) firing activity of an STN neuron, and extracellular activity of an ED-GP (4) and a P-GP (5) neuron. (B) Putative linear phase histograms (extrapolated from firing activity shown in A) for STN (blue, top), putative prototypic P-GP (red, middle), and putative arkypallidal ED-GP (green, bottom) neurons during spike-wave discharge. For clarity, two cortical SWD oscillations cycles are shown, and to facilitate the comparison, the representative spike activity of each cell type (as shown in A) was superimposed below the extrapolated phase histogram and aligned to the through (dashed line) of the SWD oscillations. *Figure panels A is extracted from J.T. Paz, J.-M. Deniau, S. Charpier, Rhythmic bursting in the cortico-subthalamo-pallidal network during spontaneous genetically determined spike and wave discharges. J. Neurosci. 25 (2005) 2092–2101. http://www.ncbi.nlm.nih.gov/pubmed/15728849 (Accessed 2 December, 2014) with permission from the relevant source.*

(3 out of 13 cells). These neurons were characterized by slow, irregular firing outside the SWD episode but became rhythmically entrained during SWDs, aligning their activity with the excitation of STN neurons (Fig. 17.6A). In contrast, the remaining ~75% of GPe neurons (10 out of 13 neurons) were classified as "pausing" (P-GP), exhibiting tonic firing during interictal periods but switching to a pause with rebound burst pattern during the SWDs. Notably, the pause in P-GP coincided with increased STN firing activity (Fig. 17.6A). Remarkably, these ED-GP and P-GP neurons closely resemble arkypallidal and prototypic neurons, both in their relative proportions, their mean firing rate and firing pattern: irregular and bursty for arkypallidal-like (ED-GP) neurons and regular and tonic for prototypic-like (P-GP) neurons. Even more striking is the fact that the spike-timing phase relationship among STN, ED-GP (putative arkypallidal), and P-GP (putative prototypic) neurons during SWDs closely mirror those observed during parkinsonian beta oscillations. Specifically, the antiphase activity between prototypic and STN neurons in both oscillatory states suggests that their spike interaction during excessive synchronization is driven by prototypic inhibitory activity, rather than by STN excitatory input (Fig. 17.5C vs. Fig. 17.6B).

The origin of this rhythmic inhibition to prototypic neurons during beta oscillations and SWDs remains unclear, though, as mentioned earlier, striatopallidal D2-SPNs neurons are a strong candidate as they provide the main inhibitory input to the GPe. However, during SWDs recorded under anesthesia, striatal neurons reduce their firing rate by ~79%, due to an increased inhibition from striatal GABergic interneurons [93]. This finding was initially interpreted as evidence against striatal D2-SPNs being the primary driver of such rhythmic inhibition to prototypic neurons during SWDs. That being said, in freely moving rats, putative striatal neurons can be entrained by high-voltage spindles (HVS), the electrophysiological equivalent of SWDs. However, their firing pattern during the oscillation is characterized by single spikes that do not occur in every oscillation cycle—unlike striatal GABAergic interneurons, which exhibit more consistent oscillatory entrainment [94]. Importantly, the phase-locked activity of striatal neurons during HVS occurs just after the trough of the EEG oscillations, aligning precisely with the timing of prototypic neuron inhibition. In summary, these findings strongly suggest that the striatal D2-SPNs neurons may directly propagate synchronized oscillatory activity to the GPe in

both SWDs and beta oscillations. However, unlike in STN or GPe, where oscillatory firing is present at the single-neuron level (e.g., individual neurons can fire one action potential per oscillation cycle), oscillatory activity in D2-SPNs appears to be encoded at the population level. That is, individual D2-SPNs tend to fire single spike but not on every cycle [94]. Similarly, during beta oscillations, aberrant phase-locked firing has been shown to be selectively restricted to striatopallidal D2-SPNs neurons [16]. However, their entrainment occurs only on a minority of beta cycles, further supporting that their contribution to driving pathological oscillatory dynamics arises through population-level entrainment. The idea that GPe activity is primarily controlled by striatal inhibition during excessive neuronal synchronization—rather than excitatory inputs from STN—is supported by anatomical evidence. On average, if each D2-SPNs in the rat striatum forms approximately 224 synapses within the GPe [95], while each STN neuron forms about 457 synapses [96], and considering that the total number of D2-SPNs is estimated to be around 1.4 million, compared with the approximately 13,600 STN neurons and 46,000 GPe [97], then the cumulative number of striatal D2-SPNs synapses formed onto single GPe neuron far outweighs those from STN. This results in an estimated striatal-to-STN synaptic ratios ranging from 30:1 to 50:1.

6. Conclusion on the phase propagation dynamics during synchronized oscillatory activity

In this chapter, we challenge the prevailing view that the STN is the primary driver of parkinsonian beta oscillations in the BG circuits of PD rat model. Drawing on data from 6-OHDA rodent models, we present evidence that GABAergic inhibition from the GPe plays a central role in generating and coordinating these synchronous beta oscillations. We propose that the GPe acts as a key hub within the BG, orchestrating the precise spike-timing relationships observed within and across BG neuronal populations. This framework builds upon earlier claims that emphasized the role of striatopallidal mechanisms in propagating abnormal low-frequency synchronization (<2 Hz) characteristic of the "slow-wave" cortical state [98]. We extend this perspective to account for the emergence of excessive beta oscillations present following dopamine depletion in awake or activated brain state (i.e., those lacking slow-wave activity), as well as other forms of pathological synchronization, such as the SWDs observed in absence epilepsy. The consistent phase relationships seen in both PD-related beta oscillations and SWDs, despite their differing frequencies and underlying mechanisms, suggest a fundamental constraint imposed by the intrinsic anatomical and functional connectivity of BG network.

In conclusion, analyzing the spike-timing phase relationships between neuronal population during synchronized oscillations serves as a powerful tool for dissecting the contributions of different neuronal element to the chrono-circuitry of brain oscillations. We propose that applying such phase-based analyses to simultaneously recorded STN and GPe neurons in nonhuman primates could yield valuable insights into whether similar circuit mechanisms underlie beta oscillations generation in MPTP monkeys as in rodent PD models. Overall, this framework suggests that the temporal dynamics of neuronal circuits are shaped by the network's intrinsic architecture and functional connectivity, which together determine when and how different neuronal populations become entrained to ongoing oscillatory activity.

Declaration of interests

The authors declare no competing interests.

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